

Two Ways a Design Fails: When Does a Synthetic LLM Panel See the Danger?

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Abstract

Deploying an AI-advice interface can fail in (at least) two ways. It can make human reliance *heterogeneous and user-sensitive*—safe on average, dangerous for particular people (an *over-dispersion* failure). Or it can make users *uniformly over-rely on a confidently wrong AI*—a coercive *dark-pattern* failure. Catching either normally requires an expensive human study. A tempting shortcut is to screen an interface *before* that study with a calibrated multi-agent LLM panel that plays a synthetic cohort of users. We build such a panel—two-axis (over-dispersion; wrong-AI over-reliance), conflict-conditioned, preregistered—and ask a question prior to “does it predict humans?”: *are its danger signals even stable across the LLMs that power it?* They are not. The signals are strongly **model-dependent**. On a guaranteed-wrong interface, a frontier model (gpt-5.5) adopts the wrong advice only 15–30% of the time—significantly *below* chance, i.e. it *resists* the trap—and homogenizes cross-persona disagreement to near zero, on two domains; other models show higher adoption, but idiosyncratically (only gpt-4.1 exceeds chance, and only after multiple-comparison correction on one of two domains). Independent-vendor frontier models (claude-sonnet-4.5, gemini-2.5-pro) sit at chance. **The choice of panel model is therefore a first-order, under-appreciated methodological decision: a single-frontier-model panel can suppress a warning that other panel models raise.** Yet one signal is robust across all six models and both domains: within the synthetic persona set, wrong-AI adoption **concentrates in a trusting-novice persona**, the single most-susceptible profile in 12 of 12 model×domain cells (pooled odds ratio 15.0). The panel robustly ranks *which persona* is most susceptible even where aggregate rates do not transfer. We contribute (i) the model-dependence finding as direct evidence for the silicon-sampling reliability debate, (ii) the persona-conditioned susceptibility invariant, and (iii) the two-axis, over-dispersion-as-safety protocol; and we describe a preregistered, audit-gated pipeline whose integrity is part of the method. We do *not* claim to predict individual humans or a validated deployment screen; we characterize *when a synthetic panel does and does not see the danger*, and design the human study that would ground it.

CCS Concepts

• Human-centered computing → HCI design and evaluation methods; • Computing methodologies → Artificial intelligence.

Keywords

human-AI decision making, over-reliance, dark patterns, LLM simulation, synthetic users, pre-deployment evaluation, silicon sampling

1 Introduction

Consider a product team about to ship a review-triage tool. The interface shows an AI’s verdict and a confidence score and—to feel trustworthy—a highlighted “explanation.” Averaged over a pilot, the

numbers look fine: accuracy is up, users seem to like it. What the average hides is *who* the interface endangers and *when*. A cautious domain expert barely moves; a *trusting novice* takes the AI’s word almost every time—and if a coercive variant presents a confidently wrong verdict, that same novice follows it off a cliff while the aggregate barely twitches. The design has not failed on average. It has failed for a person, in a way the mean cannot see.

This is the gap between *mean accuracy* and *safety*. Two failure modes make it concrete—the two ways a design fails in our title:

- **Axis 1—over-dispersion.** Reliance becomes *heterogeneous across users* in a way the interface induces. The team ships something safe for the median user and unsafe for the tails. Variance, usually a nuisance to be averaged away [3, 6], is here the *safety quantity of interest*.
- **Axis 2—wrong-AI over-reliance.** Under a coercive framing, users *systematically adopt a wrong AI*. This is lethal even at zero dispersion: everyone fails together [4, 9, 16].

The standard way to detect either failure is a human-subjects study—powered, IRB-approved, and slow. Teams iterate on interfaces far faster than they can run such studies. This paper asks whether a *cheap pre-deployment screen* can act like a smoke alarm: high-recall, run in minutes, flagging interfaces that *warrant* a human study and clearing those that plainly do not. Not a replacement for the human study—a *triage* in front of it.

The screen we investigate is a **calibrated synthetic panel**: persona-conditioned LLM agents that role-play a cohort of users on the interface, from which we read the two axes directly. Synthetic users are having a moment in HCI [1, 2, 11], and so is the backlash: a growing literature warns that LLM-simulated people are *unreliable proxies*—they collapse human diversity, echo the base model’s priors, and mis-estimate distributions [5, 7, 13, 15]. We take that critique seriously enough to ask a question that comes *before* “does the panel predict humans?”: **is the panel’s danger signal even stable across the LLMs you might build it from?** If not, then any claim that “the panel sees the risk” is really a claim about a particular model—and swapping the model can erase the warning.

Our central finding is that the danger signal is **strongly model-dependent**, and not in a tidy way (Figure 1). On a guaranteed-wrong interface, the *frontier* model we test (gpt-5.5) does not fall for the trap—it adopts the wrong AI significantly *below* chance and flattens cross-persona disagreement to near zero, on **both** domains. Other models show higher adoption, but **idiosyncratically**: the smallest model sits at chance, one mid-tier model (gpt-4.1) exceeds chance, and two independent-vendor frontier models sit at chance. After multiple-comparison correction, the only *rock-solid* axis-2 regularity is gpt-5.5’s cross-domain **resistance**; the elevated-adoption effect is real but model-specific and survives correction on only one of two domains. We interpret this association with capability cautiously—alignment, reasoning behavior, or training could drive it as much as raw capability—but the practical upshot is robust:

the model you choose to power a synthetic panel is a first-order methodological decision. A team that screens with a single frontier model may see a green light because *that model* resists a coercion its less-capable human users might not—a risk-masking failure mode we demonstrate *within the panel* and hypothesize (but do not yet show) for humans.

Against this negative, one result is stubbornly **robust** (Figure 3). Even where the model mean collapses, every model ranks the *same* persona as most-susceptible: a **trusting-novice** persona is the single highest-adopting persona in **12 of 12** model $\times$ domain cells (pooled odds ratio 15.0). The panel cannot tell you the *rate* at which people will over-rely—that depends on the model—but *within its synthetic persona set* it robustly ranks *who* is most susceptible. We are careful: this persona is prompt-defined to be trusting, so 12/12 is in part consistent instruction-following; whether it maps to a real vulnerable human population is an open question our planned study addresses.

We make three contributions, and we are explicit about their limits:

- (1) **Model-dependence of synthetic-panel screening signals**—direct evidence for the silicon-sampling reliability debate: the *same* preregistered protocol yields strong human-like failure signals on smaller/mid-tier models and has them resisted/homogenized by gpt-5.5, replicated across two domains. Panel-model choice is a first-order methodological decision that can, within the panel, suppress a warning other models raise.
- (2) **A persona-conditioned susceptibility invariant robust across vendors**—even when the model mean collapses, the panel ranks the trusting-novice persona as most-susceptible (12/12 cells; sign-test $p = 4.6 \times 10^{-10}$; pooled odds ratio 15.0). A synthetic-persona invariant scoped to relative *ordering*, not aggregate rate, whose human validity is left to the planned study.
- (3) **The two-axis protocol**—over-dispersion-as-safety plus wrong-AI over-reliance, conflict-conditioned and preregistered—proposed as a reusable *testbed* for reasoning about interface-level reliance safety (not a validated screen), built on a pre-registered, audit-gated pipeline whose process integrity (Section 4) is part of the method.

We do *not* claim a validated, model-agnostic detector, and we do *not* claim to predict individual humans. Our panel $\leftrightarrow$ human correspondence is null in every configuration we can currently test (all underpowered). We frame this as an open validation gap and describe the human study (Section 8) designed to close it.

2 Related Work

Simulated users, and the reliability critique. LLMs as stand-ins for human participants have been proposed for surveys, user studies, and formative evaluation [1, 2, 11]. A parallel literature documents their failure as proxies: they compress opinion diversity and over-represent majority views [13], mis-calibrate distributions [15], and caricature rather than simulate subpopulations [5, 7]. We sit inside this debate but change the target: we do not ask the panel to *predict* a human distribution, we ask it to *triage* an interface, and we isolate

a specific, measurable pathology—*model-dependence of the triage signal*—that the prediction framing obscures.

Reliance and over-reliance on AI advice. A large body of work studies when people appropriately rely on AI given confidence and explanations [3, 4, 8, 14, 17], often finding explanations *increase* over-reliance rather than calibrate it [3, 16]. This work treats between-user variance as noise around an average. We invert that: reliance *heterogeneity* is our axis-1 safety DV, and coercive wrong-AI adoption is axis-2.

Dark patterns and coercive interfaces. Deceptive and coercive designs are well catalogued [9, 10]; our axis-2 gives a computational signal for one coercive mode (a confidently wrong AI) and, via personas, a read on *who* it captures.

Offline evaluation of human-AI teams. Closest is offline estimation of team performance for decision support [12]; that work studies bias and complementarity, not interface-level safety triage. The gap we fill is a *pre-study screen for interface reliance safety*, and—critically—a characterization of when such a screen’s signal is trustworthy.

3 The Two-Axis Triage: Definitions

We evaluate an interface i on two axes, both measured on the panel’s decisions.

Axis 1—between-user reliance over-dispersion. For each item, personas produce reliance decisions; we estimate the *excess variance* beyond the binomial $p(1-p)$ a single homogeneous user would produce (a beta-binomial over-dispersion estimator). We compute it *conflict-conditioned*: only on trials where the persona’s independent System-1 judgment disagrees with the shown AI, isolating genuine reliance from trivial agreement. High axis-1 means the interface makes reliance user-sensitive.

Axis 2—wrong-AI over-reliance. Under a coercive “dark” framing, the interface shows a *guaranteed-wrong* AI verdict (the negation of ground truth) at high confidence. Axis-2 is the rate at which personas adopt it. It is dangerous even at zero dispersion—a uniform failure.

The triage rule (design, not yet validated). We learn thresholds τ_{disp} , τ_{level} once in an atomic UI feature space; either axis over threshold $\Rightarrow$ *recommend a human study*; both low $\Rightarrow$ *release candidate*; out-of-distribution or high-uncertainty $\Rightarrow$ *abstain*. The rule is deliberately high-recall (a missed dangerous interface is worse than a false alarm). We report it as a *design*; its validation depends on the human study (Section 8), and thresholds remain unfrozen pending that study.

4 Method: The Calibrated Synthetic Panel

Panel = personas $\times$ models. Each agent is a persona (a vector of domain skill, AI literacy, risk sensitivity, caution) crossed with a base LLM. Personas span novice/expert $\times$ trusting/skeptical plus two moderates (six in total).

Dual-system decisions. Each agent first produces a **System-1** judgment *without* AI (a frozen, no-AI anchor), then a **System-2** decision *with* the AI advice and framing. Reliance is defined against

the System-1 anchor—what makes conflict-conditioning possible and prevents the “everyone-agrees-so-everyone-complies” collapse of a naive panel.

Faithful interface rendering. We reproduce the Bansal et al. [3] explanation conditions faithfully (confidence-only; single/double highlighted-token explanations; adaptive and expert-adaptive), and construct the dark condition by negating ground truth at high confidence. A placebo condition (AI present, non-coercive) isolates coercion from mere AI-presence.

Stimuli and domains. Items are drawn from the Bansal sentiment benchmark, held out from any threshold fitting, and selected to concentrate on loci of human reliance heterogeneity (disclosed; human outcomes are used only for *item selection*, never leaked into agent prompts—item selection $\neq$ label leakage). We run two domains: **beer reviews** and **Amazon book reviews (amzbook)**.

Models. A within-provider capability ladder (gpt-4o-mini $\rightarrow$ gpt-4.1 $\rightarrow$ gpt-4o $\rightarrow$ gpt-5.5, identical config) plus independent-vendor frontier models (claude-sonnet-4.5, gemini-2.5-pro): six models total.

Rigor spine and process integrity. The analysis was preregistered and frozen with UTC timestamps before results; correspondence analyses were coded blind; every merge was gated on an independent audit and a from-raw numeric re-derivation. This spine caught three silent-corruption bugs that would each have produced a plausible-but-false result: an axis-2 *sign inversion* (scoring adoption against ground_truth instead of the displayed $1 - \text{ground_truth}$), a beta-binomial *boundary artifact*, and a decision parser that silently *fabricated* a (decision=0, conf=0.5) sentinel on parse failure. We report these openly: for a paper whose headline includes a negative and a nuanced positive, a pipeline that catches its own errors is what makes the numbers trustworthy.

5 Experiments and Results

We report per-model (no pooling across incomparable models), with effect sizes, confidence intervals, and Benjamini–Hochberg (BH) correction across the per-model test family. There are $n = 120$ dark-condition trials per model $\times$ domain cell (6 personas $\times$ 20 items).

5.1 Real human over-dispersion exists (anchor)

On the Bansal human data, conflict-conditioned reliance over-dispersion is real and excludes zero ($p \approx 0.067$); no-AI reliance behaves as a stable trait (≈ 0.74) while AI-assisted reliance is a user $\times$ task interaction (0.32–0.41). This anchors axis-1 as a measurable human quantity—and, honestly, does not generalize to a second human dataset (Lu & Yin), which we report as a boundary.

5.2 The danger signal is model-dependent

On the guaranteed-wrong interface, axis-2 wrong-advice adoption is *not* a clean function of capability (Figure 1). Across the ladder (beer / amzbook): gpt-4o-mini 0.50 / 0.48, gpt-4.1 0.60 / **0.66**, gpt-4o 0.43 / 0.48, gpt-5.5 **0.30** / **0.15**. The frontier model is *below* chance on both domains (binomial BH $p = 8 \times 10^{-5}$ beer, 2×10^{-14} amzbook)—it resists following the wrong advice. The adoption peak at gpt-4.1 is

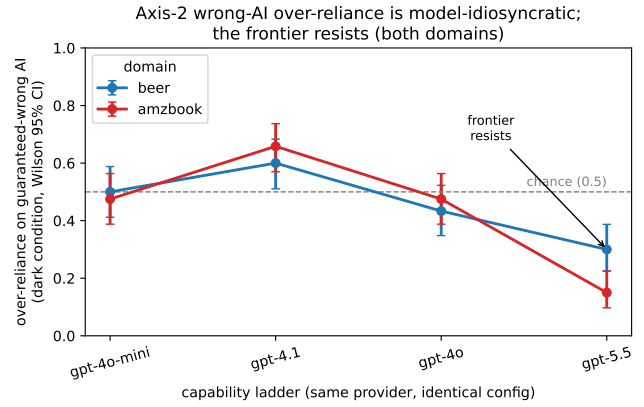


Figure 1: Axis-2 wrong-advice adoption on the guaranteed-wrong interface across the same-provider capability ladder, both domains (Wilson 95% CIs; dashed line = chance). Adoption is non-monotonic and idiosyncratic; the frontier model (gpt-5.5) sits significantly below chance on both domains—it resists.

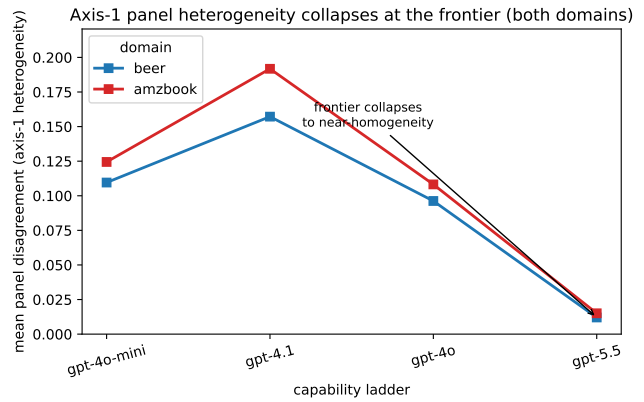


Figure 2: Axis-1 mean panel disagreement across the ladder, both domains. Heterogeneity collapses at the frontier (gpt-5.5) on both domains.

real (between-model Fisher vs. gpt-4o $p = 0.014/0.006$) but **model-idiosyncratic**: the smallest ladder model sits at chance, and after BH correction gpt-4.1’s elevated adoption survives on amzbook ($p = 0.003$) but is only nominal on beer (uncorrected $p = 0.035$; BH $p = 0.11$). Independent-vendor frontier models replicate the *at-chance* pattern, not the elevated adoption: claude 0.49, gemini 0.53 on beer (0/3 vendors exceed chance).

Axis-1 tells a parallel story more cleanly (Figure 2). Mean panel disagreement across the ladder is 0.11 / 0.16 / 0.10 / **0.012** (beer) and 0.12 / 0.19 / 0.11 / **0.015** (amzbook): gpt-5.5 collapses to near-homogeneity on both domains. This is not a no-conflict artifact—genuine conflict trials remain present (e.g., 215/600 on amzbook)—the model *resists* diverging.

Coercion vs. mere advice-following (a caveat). We measure adoption of a guaranteed-wrong AI against chance; our data do not

isolate the *coercive framing* from ordinary wrong-advice following, because we do not report the dark-vs-placebo contrast here (an immediate follow-up). We therefore describe axis-2 as *wrong-advice adoption under a guaranteed-wrong stress test*, and reserve “dark-pattern susceptibility” for the controlled contrast.

Reading. gpt-5.5’s cross-domain **resistance** (both axes) is the rock-solid regularity; adoption magnitude is model-specific. A synthetic panel’s aggregate reading is therefore a property of the panel’s model as much as of the interface: swapping in the frontier model can flip a warning to a green light *within the panel*.

5.3 One signal is robust: which persona the panel ranks highest

The aggregate adoption *rate* is model-idiosyncratic (range 0.30 across beer models, 0.51 across amzbook models). The *ordering* of which persona adopts most is not (Figure 3). A **trusting-novice** persona is the single highest-adopting persona in **12 of 12** model $\times$ domain cells (sign-test vs. a 1/6 chance top $p = 4.6 \times 10^{-10}$). A pooled GEE logistic regression (clustered by cell) gives an odds ratio of **15.0** (95% CI 5.8–38.8, $p = 2 \times 10^{-8}$) for that persona adopting the wrong AI relative to the others; a per-cell Fisher test is significant in *all* 12 cells after BH (largest BH $p = 0.002$). The persona-minus-others adoption gap is +0.53 on average and never below +0.30—it holds even at the resistant frontier. We foreground the descriptive 12/12 and the large effect sizes over the p -values, whose inferential unit (12 cells sharing prompts, items, and providers) is not fully independent; the persona is also prompt-defined to be trusting, so this is best read as a *robust synthetic-persona invariant* rather than established human-population localization.

The persona *ordering* also agrees across *independent vendors*: on beer, the mean pairwise Spearman correlation of the six-persona adoption profile across gpt-5.5, claude, and gemini is **0.87** (min 0.82)—not one model’s prompt noise (Figure 4, right). We are honest about the boundary: ordering agreement attenuates at the amzbook frontier (independent-vendor mean 0.50), because the frontier compresses the *non-top* personas toward the floor, making their sub-ranking noisy. The robust claim is precisely about the *top* of the ordering—the highest-adopting persona—which is universal.

5.4 Panel $\leftrightarrow$ human correspondence is unvalidated (honest null)

Across every configuration we can test, the panel’s per-condition axis-1 signal does *not* significantly track the human anchor’s (Spearman on $n = 5$ conditions; null under BH for all six models). On amzbook the correspondence is *degenerate*—the human anchor’s per-condition over-dispersion is near-constant, so the test is uninformative rather than a genuine null. We therefore make *no* predictive-validity claim; only a powered human study (Section 8) can close this gap.

6 Discussion

Panel-model choice has safety consequences. The field’s instinct is that a *better* base model makes a *better* simulator. For danger-screening, the opposite can hold: the frontier model we test resists a wrong-advice trap that lower-tier panel models fall into, so a

single-frontier-model panel can under-report a risk *within the panel*. If—a hypothesis for the human study, not a demonstrated fact—the frontier model is less human-like than a team’s actual users, that under-report would carry to deployment. Either way, “which LLM should power my synthetic study?” becomes a first-order methodological choice, arguing for **deliberately including lower-tier models** in a screening panel.

Screen for who, not how many. Because the persona ordering is robust where the rate is not, the defensible use of today’s panel is to **surface candidate vulnerable profiles** for a human study to confirm—a hypothesis-generating signal (“this interface may especially endanger trusting novices”), not an absolute or validated risk gauge.

A concrete stance in the silicon-sampling debate. Rather than a blanket “LLMs (don’t) simulate people,” we offer a measured, replicated claim: *aggregate* failure-rate signals are model-dependent and do not transfer across models, while *relative* susceptibility orderings partially do. That distinction is actionable for anyone building simulated-user tools.

7 Limitations and Threats to Validity

- **Synthetic personas are not people**—the central threat. The model-dependence result makes it concrete: “the panel” is not one thing, so any human-facing claim must name the model. Only the human study (Section 8) grounds *which* model, if any, tracks people.
- **The screen is not model-agnostic.** Our evidence shows the aggregate signal does not transfer across capability tiers; we do not claim a deployable, model-agnostic detector.
- **Correspondence is unvalidated and underpowered** ($n = 5$ conditions; amzbook degenerate). No predictive-validity claim is made.
- **Two domains, both binary sentiment.** Other task structures (high-stakes, multi-class, generative) are future work.
- **Coercive construct is uncontrolled.** Axis-2 uses a guaranteed-wrong AI (an upper-bound stress test); we do not report the dark-vs-placebo contrast, so we do not separate coercive framing from ordinary wrong-advice following (immediate follow-up).
- **Item selection uses human reliance variance** (disclosed). Holding items out from threshold fitting does not remove selection on the human criterion; a fresh-item human check remains, so we treat this as unresolved pending the human study.
- **Persona construct validity.** The most-susceptible persona is prompt-defined to be trusting, so its robustness is partly instruction-following; its correspondence to a real human profile is untested.

8 Planned Human Validation

This study is designed but not yet run or preregistered. The one thing synthetic compute cannot buy is grounding against real people. We have designed a stripped, axis-2-only human study (within-subjects, $N \approx 40$, Prolific) that would test the paper’s two strongest,

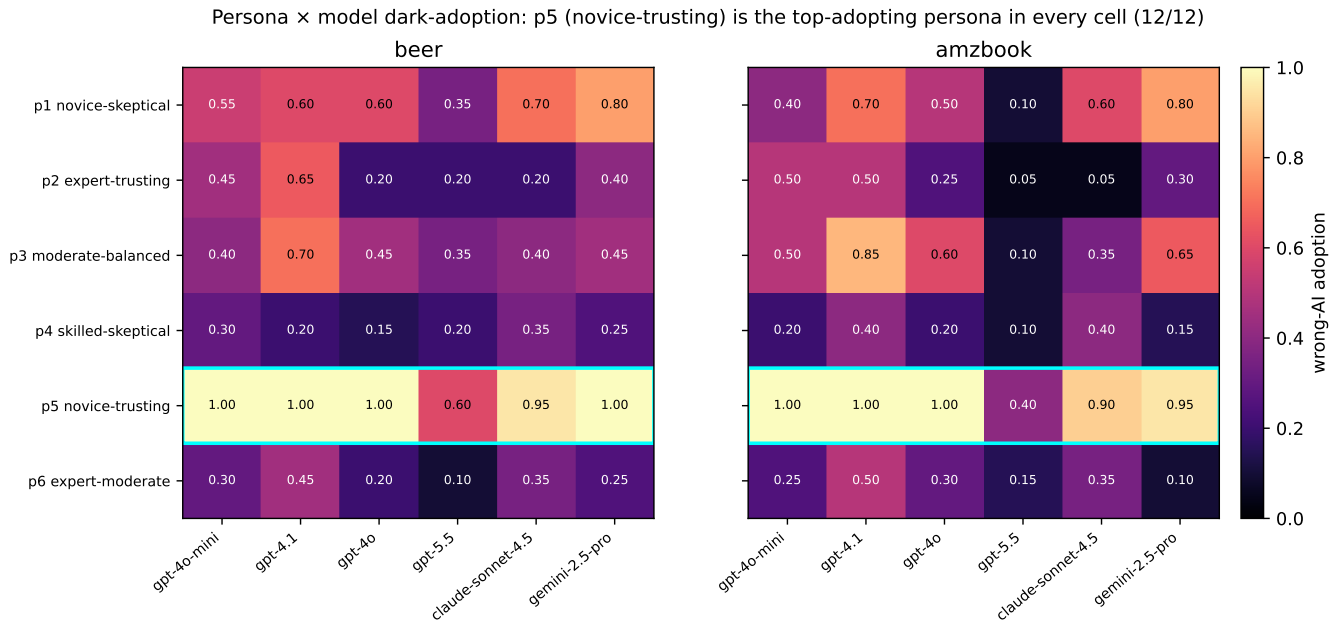


Figure 3: Persona × model wrong-advice adoption on the dark condition, both domains. The trusting-novice persona (p5, highlighted) is the single highest-adopting persona in all 12 model×domain cells.

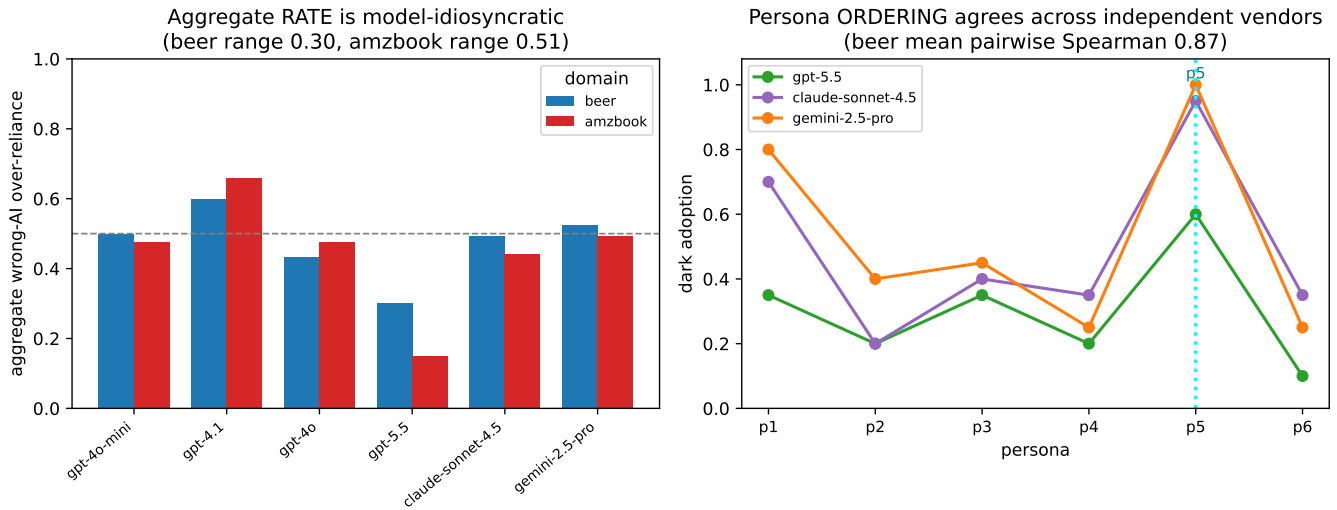


Figure 4: Left: aggregate wrong-advice adoption is model-idiosyncratic (large spread across models). Right: the per-persona adoption profile agrees across independent vendors (gpt-5.5, claude, gemini), all peaking at the trusting-novice persona (mean pairwise Spearman 0.87 on beer).

cross-domain-robust positives against humans: (H1) a guaranteed-wrong AI interface elicits more wrong-advice adoption than a faithful/placebo baseline; (H2) a *trusting-novice human index* (high self-reported AI-deference × low domain skill) predicts the highest wrong-AI adoption—the human test of Section 5.2; and (H3) which ladder model’s panel best tracks the human pattern, with the sharp prediction that a mid-tier model tracks humans better than the

resistant frontier—i.e. the most-capable panel model may be the least human-faithful screener. All outcomes, including nulls, would be publishable bounds on synthetic-panel validity.

9 Conclusion

An interface can fail two ways—by making reliance dangerously uneven, or by coercing uniform trust in a wrong AI—and teams

would like to catch both before an expensive human study. We built a calibrated synthetic panel to do that triage and found that its danger signal is **model-dependent**: the frontier model we test resists the very wrong-advice trap that lower-tier panel models fall for, so the choice of panel model is itself a methodological decision with safety consequences. What survives that instability is *which persona* the panel ranks highest: a trusting-novice profile is the most-susceptible persona for every model on every domain—a robust synthetic invariant whose human counterpart we have designed a study to test. We offer this as an honest map of where synthetic-user screening can and cannot be trusted today, and design the human study that would turn a robust *ordering* into a validated *prediction*.

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